

PLSC 197 / SODA 110N: Social Data, Technology, and Artificial Intelligence

Christopher Zorn

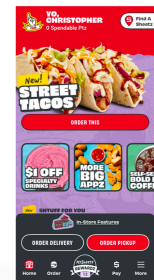
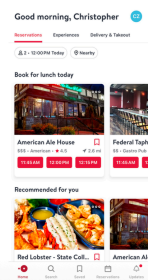
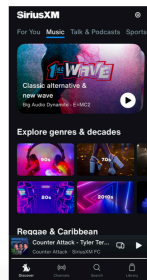
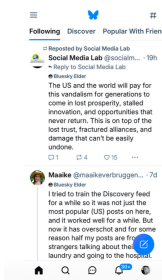
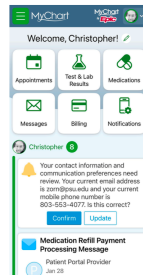
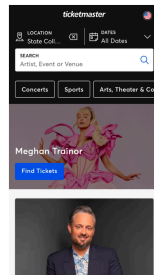
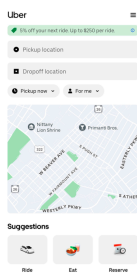
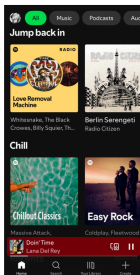
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“A digital platform is a software-based online infrastructure that facilitates interactions, transactions, and value exchange between distinct user groups (such as buyers and sellers or content creators and consumers). It acts as a central hub for business operations, utilizing technologies like APIs and data analytics to streamline processes, foster connectivity, and generate network effects.”

- Google Gemini

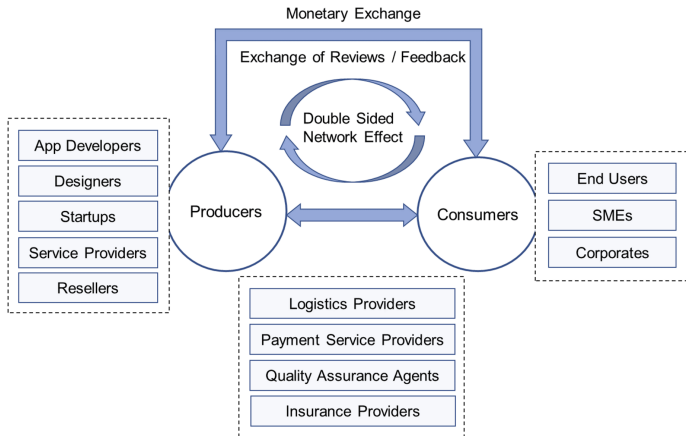
Platforms are where the data get made.

Examples...



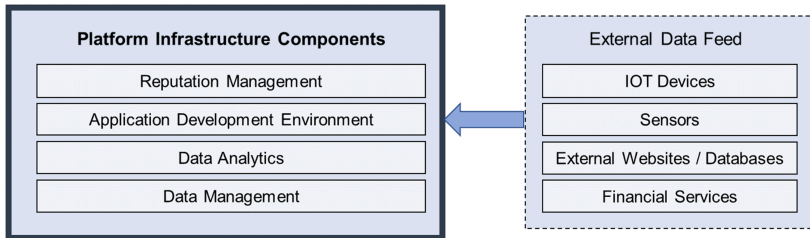
The Ecosystem

Digital Platform Ecosystem



(Source)

The Infrastructure

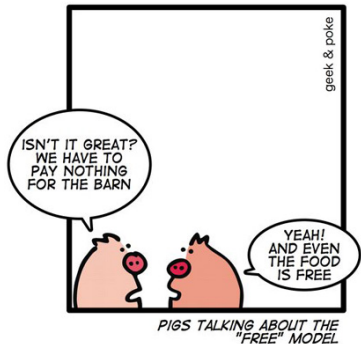


(Source)

van Dijck (2018): *Platform Mechanisms*

Three mechanisms:

- *Datafication*
- *Commodification*
- *Selection*



Datafication:

“Datafication refers to the process by which subjects, objects, and practices are transformed into digital data.”

– Southerton (2022)

What it isn't:

- Digitization
- Quantification
- “Objective”

Datafication (continued)

What makes datafication different?

- Data **Capture**
 - Ubiquity
 - Velocity
 - Networks / connection
- Data **Circulation**
 - The data are the product
 - Users are active participants

...is:

“...platforms transforming online and offline objects, activities, emotions, and ideas into tradable commodities.”

– van Dijck (2018)

“Currencies”:

- | | |
|-------------|---------|
| • Attention | • Users |
| • Data | • Money |

Commodification: Example

Apple Fitness +:

- Physical activity (steps taken, heart rate, sleep patterns, etc.) → Data
- Apple → Aetna, to:
 - ...offer wellness programs
 - ...assess risks
 - ...set premiums
- 🕒 Aetna offers free Apple Watches to members



Commodification: Effects

Platform-driven commodification:

- ...drives *multisided markets*
- ...*empowers* consumers / smaller actors
- ...fosters *monopolies*
- ...creates “*middlemen*” (e.g., data services)
- ...others?

Selection (also, “Curation”)

How platforms steer user interaction.

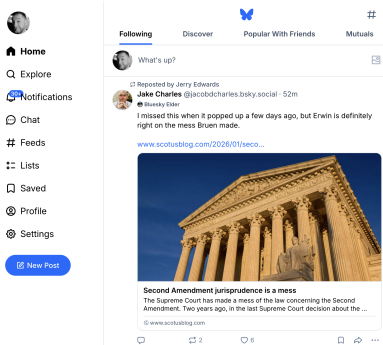
Platforms curate user content + influence user activity via:

- Personalization
- Reputation and trends
- Moderation

Personalization

Personalization:

- ...combines *individual* and *aggregate* data
- ...balances personal and popular foci (via algorithms)
- benefits: a “custom” / user-driven experience
- concerns: “filter bubbles,” algorithmic values



File under “Things platforms do to promote engagement” ...

Characteristics:

- Both are products of the data...
- Both can be platform-, service-, or user-directed
- Both are (varyingly) responsive to – and therefore drive/incentivize – user behavior
- Both are (deliberately) capable of rapid change

“...a technology-driven toolset designed to monitor, filter, and manage user-generated content across digital channels.”

It:

- Is central to social media platforms (but exists in all)
- Blends automated and human-driven
- Balances desire for openness with need to avoid reputational + legal costs, and so
- Embodies the (contested) values of the platform owners/operators

Summary: Platform Mechanisms

Comparison of Platform and Non-Platform Mechanisms

Phenomenon	Non-Platform Mechanism	Platform Mechanism
<u>Datafication</u>		
Data Collection	Narrow, "slow," intentional/active	Ubiquitous, constant, passive
Data Circulation	Closely-held, users are uninvolved	Data = product, users are active participants
<u>Commodification</u>		
Data Commodification	Currencies: users + money, unidirectional	Add attention + data, "feedback loops"
Nature of the Market	One-sided (e.g., Lowes)	Often multi-sided (e.g., AirBnB)
<u>Selection</u>		
Selection Mechanism(s)	Experts / producers	Users and/or algorithms
Personalization	Low + bespoke + costly	High + at scale + free (or "freemium")
Reputation	Slow	Fast
Moderation	N/A / minimal	Central, contested